Xuewen Luo

Ph.D. Student, Transportation Engineering
University of Utah
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EDUCATION

University of Utah
Ph.D. in Transportation Engineering *08/2025 – Present*

- Research Focus: Autonomous Driving and Vision-Language Models (VLMs)

Monash University
Master of Artificial Intelligence *07/2023 – 07/2025*

- Honors: High Distinction
- Relevant Coursework: Computer Architecture and Networks, Machine Learning, Deep Learning, Introduction to Databases, IT Research Methods, Modeling Discrete Optimization Problems

Yunnan University
Bachelor of Logistics Management *09/2018 – 06/2022*

- Relevant Coursework: Advanced Mathematics, Fundamentals of Programming Design, Application of Lean Transformation Information Technology, Fundamentals of Computers, Logistics System Simulation

PUBLICATIONS

Journal Articles

1. F. Yang, B. Yu, Y. Zhou, **X. Luo**, Z. Tu, C. Liu. "Edge-Based Multimodal Sensor Data Fusion with Vision-Language-Action (VLA) Model for Real-Time Autonomous Vehicle Accident Avoidance." *Engineering Applications of Artificial Intelligence*, vol. 180, 115186, 2026.

Refereed Conference Proceedings

1. **X. Luo**, F. Yang, D. Fan, X. Gao, B. Yu, Z. Li, Z. Tu, Y. Zhou, C. Liu. "V2X-UniPool: Unifying Multimodal Perception and Knowledge Reasoning for Autonomous Driving." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pp. 747–756, 2026.
2. F. Ding, H. H. Tew, J. Y. Loo, L. T. Liu, F. Y. Leong, **X. Luo**, K. K. Chin, J. J. Gan. "GSMT: Graph Fusion and Spatiotemporal Task Correction for Multi-Bus Trajectory Prediction." *IEEE 104th Vehicular Technology Conference (VTC2026-Fall)*, 2026.
3. S. You, **X. Luo**, X. Liang, J. Yu, C. Zheng, J. Gong. "A Comprehensive LLM-Powered Framework for Driving Intelligence Evaluation." *IEEE International Conference on Robotics and Automation (ICRA)*, pp. 7028–7035, 2025.
4. **X. Luo**, C. Liu, F. Ding, F. Yang, Y. Zhou, J. Y. Loo, H. H. Tew. "SenseRAG: Constructing Environmental Knowledge Bases with Proactive Querying for LLM-Based Autonomous Driving." *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, pp. 989–996, 2025.
5. **X. Luo**, F. Ding, R. Chen, R. Panda, J. Y. Loo, S. Zhang. "What's Happening: A Human-Centered Multimodal Interpreter Explaining the Actions of Autonomous Vehicles." *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, pp. 1073–1080, 2025.
6. **X. Luo**, F. Ding, Y. Song, X. Zhang, J. Y. Loo. "PKRD-CoT: A Unified Chain-of-Thought Prompting for Multi-Modal Large Language Models in Autonomous Driving." *International Conference on Neural Information Processing (ICONIP)*, pp. 62–76, 2024.

7. F. Ding, **X. Luo**, G. Li, H. H. Tew, J. Y. Loo, C. W. Tong, A. S. M. Bakibillah, Z. Zhao, Z. Tao. "Energy-Efficient Hybrid Model Predictive Trajectory Planning for Autonomous Electric Vehicles." *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 1263–1269, 2024.
8. J. Q. Loh, **X. Luo**, F. Ding, H. H. Tew, J. Y. Loo, Z. Y. Ding, S. Susilawati, C. P. Tan. "Cross-Domain Transfer Learning using Attention Latent Features for Multi-Agent Trajectory Prediction." *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 3905–3911, 2024.
9. H. H. Tew, G. Li, F. Ding, **X. Luo**, J. Y. Loo, C. M. Ting, Z. Y. Ding, C. P. Tan. "KANS: Knowledge Discovery Graph Attention Network for Soft Sensing in Multivariate Industrial Processes." *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 4377–4383, 2024.
10. G. Li, C. H. Lim, Q. Ma, X. Tang, H. H. Tew, F. Ding, **X. Luo**. "FedBChain: A Blockchain-Enabled Federated Learning Framework for Improving DeepConvLSTM with Comparative Strategy Insights." *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 2607–2612, 2024.

Under Review / Preprints

1. **X. Luo**, L. Jiang, B. Yu, F. Ding, X. Zuo, Y. Wang. "Which Trajectory Is Safer? Combining Rule and Learning for Safety Evaluation." *Under review at the IEEE International Conference on Robotics and Automation (ICRA)*, 2027.
2. D. Fan, F. Ke, K. Y. Ma, **X. Luo**, W. Liang, V. M. Baskaran, H. Rezatofighi, Y. Wu. "JARVIS: From Visual Manuals to Autonomous Robotic Assembly." *Under review at the International Conference on Learning Representations (ICLR)*, 2027.
3. **X. Luo**, B. Yu, Y. Zhang. "From Sensors to Sentences: A Language-Driven Framework for V2X Cooperation." *Under review*, 2026.
4. **X. Luo**, D. Fan, R. Chen, Y. Cao, X. Liu, B. Yu, F. Yang, C. Liu. "Who Responds When the Driver Is Gone? A Framework for Human Intent Understanding." *arXiv preprint* arXiv:2607.04670, 2026.
5. F. Yang, B. Yu, **X. Luo**, C. Liu, C. Liu. "CIVIC: End-to-End Sequence Compactness for Efficient Vision-Language Models." *arXiv preprint* arXiv:2605.28115, 2026.
6. B. Yu, F. Yang, Y. Liu, C. Wang, **X. Luo**, T. Li, R. Ke, X. Zhou, C. Liu. "Locatability-Guided Adaptive Reasoning for Image Geo-Localization with Vision-Language Models." *arXiv preprint* arXiv:2603.13628, 2026.
7. F. Ding, **X. Luo**, F. Yang, B. Yu, H. H. Tew, G. Krishnasamy, J. Y. Loo. "KnowDiffuser: A Knowledge-Guided Diffusion Planner with LLM Reasoning." *arXiv preprint* arXiv:2603.10441, 2026.
8. F. Ding, **X. Luo**, H. H. Tew, R. Reddy, X. Wang, J. Y. Loo. "Drive As You Like: Strategy-Level Motion Planning Based on a Multi-Head Diffusion Model." *arXiv preprint* arXiv:2508.16947, 2025.
9. X. Gao, Y. Wu, F. Yang, **X. Luo**, K. Wu, X. Chen, Y. Wang, C. Liu, Y. Zhou, Z. Tu. "AirV2X: Unified Air-Ground Vehicle-to-Everything Collaboration." *arXiv preprint* arXiv:2506.19283, 2025.

RESEARCH EXPERIENCE

Research on End-to-End Autonomous Driving — Monash University
Research Intern (Supervisor: Junnyong Loo) — *10/2023 – 06/2025*

- Conducted literature reviews on autonomous driving, deep learning, and computer vision.
- Reproduced BEVFormer and BEVFusion models, performing simulations in MATLAB and PyTorch.
- Published work: *Energy-Efficient Hybrid Model Predictive Trajectory Planning for Autonomous Electric Vehicles* (SMC 2024).

Research on Smart Transportation Systems for Intersection Safety — University of Utah
Research Intern (Supervisor: Chenxi Liu) — *08/2024 – 01/2025*

- Trained YOLO models for object detection and implemented object tracking using StrongSORT.
- Developed predictive models for intersection collision scenarios using real-time traffic data.
- Published work: *SenseRAG* (WACVW 2025) and *V2X-UniPool* (CVPRW 2026).

Research on Driving Behavior Evaluation Tsinghua University
Research Intern (Supervisor: Jiangtao Gong) *05/2024 – 10/2024*

- Processed predictive driving behavior data and created knowledge graphs for analysis.
- Utilized LLM-based retrieval-augmented generation with knowledge graphs (RAG-KG) to comprehensively evaluate driving behavior.
- Published work: *LLM-Powered Framework for Driving Intelligence Evaluation* (ICRA 2025).

INDUSTRY EXPERIENCE

ML Perception Intern Glydways, San Francisco Bay Area, CA, USA
Intern *08/2026 – 12/2026*

- Working with the ML Perception team on the in-cabin project, focusing on understanding and responding to passengers inside the vehicle.
- Applying research on human-in-the-loop autonomous driving to real-world in-cabin perception problems.
- Gaining hands-on experience with perception systems in an industry setting.

LLM Algorithm Intern Li Auto, Beijing, China
Intern *12/2024 – 03/2025*

- Synthesized Chain-of-Thought (CoT) text data to enhance the reasoning capabilities of large language models.
- Investigated effective strategies for supervised fine-tuning (SFT), such as multi-agent model integration.
- Conducted fine-tuning experiments, analyzed results, and identified performance improvements.

HONORS & AWARDS

WTS Northern Utah Chapter Undergraduate/Graduate Scholarship 05/2026
WTS International, Northern Utah Chapter

IEEE ITSS Fellowship 04/2025
Fellowship Program for Promoting Women, Young Professionals, and Leadership in ITS

PROFESSIONAL SERVICES

Transportation Research Board (TRB)
Friend Member, Connected and Automated Vehicle Systems Committee
Friend Member, Integrated Transportation Safety Management Committee
American Society of Civil Engineers (ASCE)
Corresponding Member, Data Sensing & Analytics Committee
Corresponding Member, Transportation Safety Committee

REFEREE FOR JOURNALS AND CONFERENCES

Conferences
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026
Conference on Neural Information Processing Systems (NeurIPS) 2026
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2025
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2025
COTA International Conference of Transportation Professionals (CICTP) 2025
International Conference on Intelligent Computing (ICIC) 2025
Transportation Research Board (TRB) Annual Meeting 2026
Journals
Frontiers of Information Technology & Electronic Engineering